# RISK MANAGER

A Late-Fusion AI System for Detecting Empty-Box and Substitution Return Fraud in E-Commerce

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This report accompanies the GitHub repository and pitch video submitted for Razorpay's AI Buildathon 2026, Track 02: AI Risk Manager. It covers why this problem was chosen, how existing industry and academic approaches handle return fraud, what was actually built, an honest look at the results, and what sets this apart from a typical single-model submission.

The goal throughout was to build something that could hold up under real scrutiny, not just look good in a demo. Where the evaluation has real limits, mainly around how much real visual data a solo, zero-budget build can realistically gather, those limits are stated directly rather than glossed over. A result that's honestly scoped is worth more than one that looks better on paper.

## Contents:

## 1. About the Author & Motivation:

I am Jey Praveen Sivaraj, a final-year student at SRM Institute of Science and Technology (SRMIST), Chennai, and currently a Research Trainee in AI/ML at Velai Health Analytics Pvt. Ltd. My work there involves applying machine learning to real-world operational data, which shaped how I approached this buildathon: I wanted to avoid building a purely academic classifier and instead design something that reflects how a real risk or fraud team would need a system to behave - with honest metrics, graceful failure handling, and decisions that account for the financial cost of being wrong in either direction.

Razorpay's Track 02 brief stood out because it explicitly rewards this kind of engineering maturity rather than raw model accuracy - it asks for honest metrics including false-positive cost, and it disqualifies anything offense-capable. That framing matched the discipline I wanted to practice building something defensible under scrutiny, not just something that demos well.

## 2. Track Chosen & Problem Statement:

**Track 02 - "AI Risk Manager":** Stop the merchant losing money to fraud, returns, and chargebacks.

**Official requirement:** "Build a working detector, verifier or auto-responder for one class of loss, with measured precision and recall on a held-out test set." The bar set by Razorpay: "Honest metrics including false-positive cost. Strictly defense-only: anything offense-capable is disqualified."

**Loss class chosen:** Empty-box and item-substitution return fraud - a customer initiates a return, but the package sent back is empty, contains a different (often cheaper) item, or is visibly damaged in a way inconsistent with normal use. This was chosen deliberately over the brief's other example directions (chargeback evidence responder, fraud-spike detector, abuse-ring sentinel) for three reasons:

• It is visually verifiable - a photo of the returned item can directly confirm or refute the claim, which allows a genuine computer-vision component rather than a purely tabular one.

• It sits at the intersection of behavioural risk and physical evidence - account history alone cannot fully resolve it, and a photo alone cannot fully resolve it either. This makes it a natural fit for a multi-modal fusion system rather than a single model.

• India-relevant and current: Surging over the past two years, return abuse is now a leading e-commerce fraud vector, surpassing payment fraud in some 2026 reports (see Section 3).

## 3. Literature & Industry Survey:

Before designing the system, existing industry and academic approaches to return-fraud detection were surveyed to understand where genuine gaps exist and to avoid re-inventing a weaker version of something that already works well in production.

### 3.1 Industry Context:

**i.** Return and refund abuse is rapidly accelerating in Indian e-commerce. Broad industry estimates from global retail reports (e.g., NRF 2023/2024) suggest fraudulent returns account for roughly 13-15% of all returns, and commentary increasingly names policy abuse as the leading fraud threat for 2026, overtaking classic payment fraud.

**ii.** Logistics-side estimates suggest merchants without structured photo/video evidence struggle significantly with disputes. They recover only a fraction of the disputed fraudulent-return value compared to when verifiable evidence exists, directly motivating a photo-verification approach.

### 3.2 Existing Commercial Approaches:

| Approach | What it does | Relation to this project |
|---|---|---|
| Increff-style WMS video capture | Order-linked photo/video capture at point of return, used as dispute evidence. | Improves evidence quality, but is largely a capture/storage layer - does not itself score or automate a decision. |
| ReturnGO / Claimlane | Behavioural + policy-rule engines flagging serial returners and policy-violation patterns. | Similar in spirit to our tabular layer, but generally does not fuse in a visual verification signal. |
| Sift / Signifyd / Riskified | Broad fraud platforms using device fingerprinting, identity graphs, and behavioural signals - mostly payment-fraud-first. | Out of scope for a solo buildathon build (requires device/identity infrastructure); noted as a natural extension. |

| | | |
|---|---|---|
| Academic: SAM-based segmentation - DINOv2 embeddings | Zero-shot segmentation and semantic similarity, increasingly used in visual-verification and anomaly-detection research. | Directly adopted here for empty-box detection and item-substitution matching - requires no task-specific fine-tuning, which suits a zero-budget solo build. |

**Positioning:** This project is a hybrid of ReturnGO-style behavioural scoring and Increff-style photo verification, combined through a genuine fusion model rather than separate tools. As explicitly stated in Section 8, it does not claim to match production vendors operating on years of labelled data and device-graph infrastructure.

## 4. Razorpay's Requirements vs. Our Solution:

| Requirement | How this submission satisfies it |
|---|---|
| One class of loss | Strictly scoped to empty-box / item-substitution return fraud only - no drift into chargebacks or abuse rings. |
| Working detector, verifier, or auto-responder | All three: a tabular detector, a vision verifier, and a three-way decision auto-responder. |
| Measured precision & recall on a held-out test set | A genuinely held-out synthetic test split (2,000 rows, touched only once) is scored for PR-AUC, precision, and recall at a cost-optimal threshold. |
| Honest metrics including false-positive cost | An explicit cost function (Expected_Cost = FP×$C_{FP}$ + FN×$C_{FN}$) is used to choose the operating threshold, with $C_{FP}$ and $C_{FN}$ stated as back-of-envelope assumptions, not ground truths. |
| Strictly defense-only | The system only flags, routes, or requests more information. It never auto-denies, never fabricates evidence, and - notably - a fake-photo detector was deliberately not built, because training one would require generating synthetic forgeries, which the brief explicitly disqualifies. |

## 5. System Architecture:

The system uses late fusion rather than early fusion: the tabular model, the semantic similarity model, and the empty-box detector each produce an independent score, which are only combined at the very end by a small meta-learner. This was a deliberate choice - return photos are frequently missing, blurry, or delayed, and an early-fusion model that concatenates raw pixel and tabular features has no clean way to say "I did not have a usable image, discount my confidence." Late fusion allows the vision signal to degrade or disappear entirely without collapsing the whole system - this is what makes the circuit-breaker behaviour in Section 6 possible.

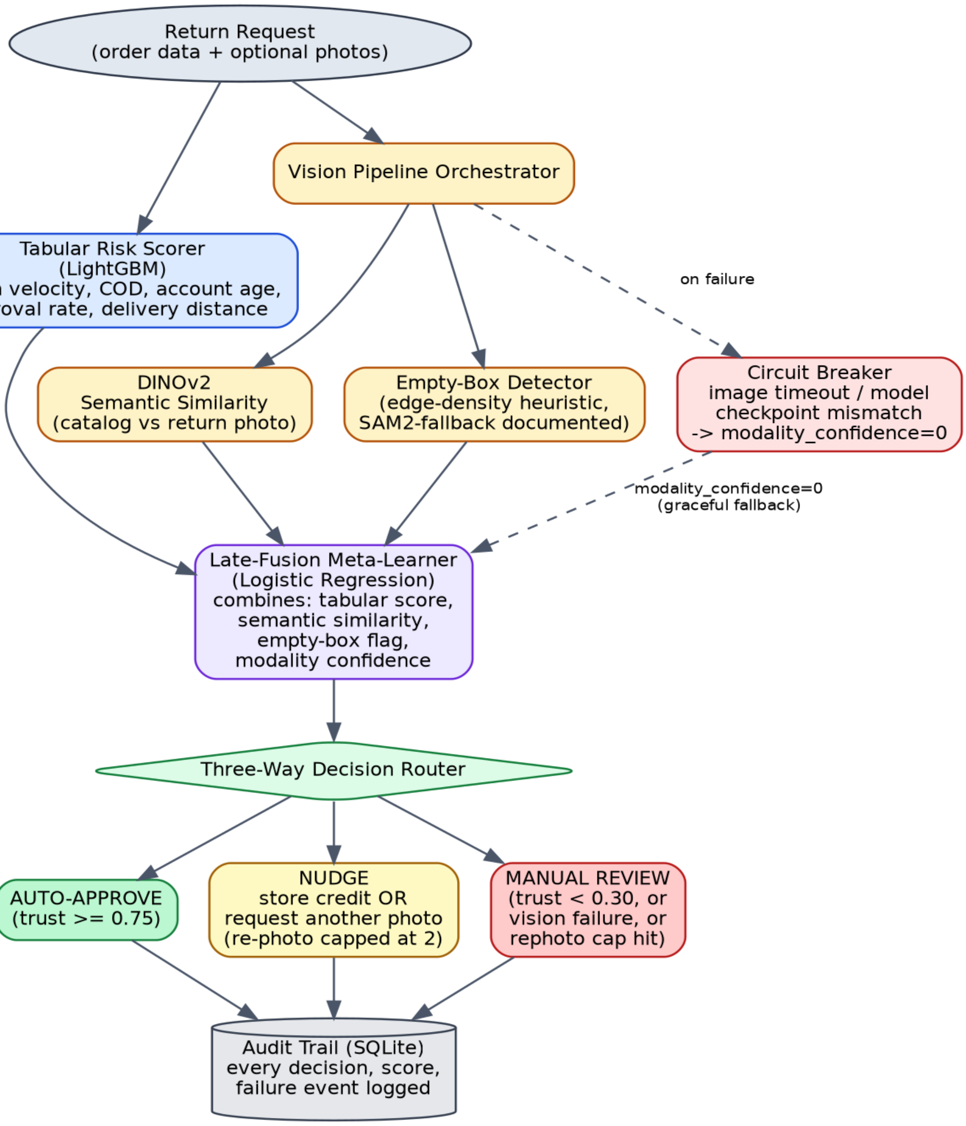


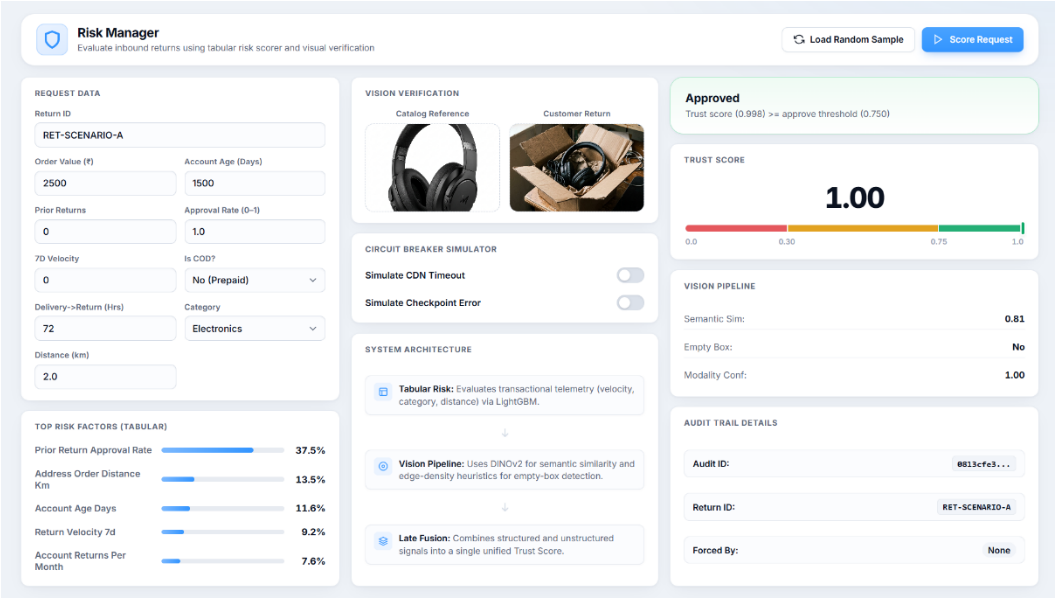

***Figure 1. End-to-end pipeline:*** *Tabular and vision signals are scored independently, fused by a logistic-regression meta-learner, and routed to one of three outcomes, with every decision written to an audit trail.*

## 6. What Was Implemented:

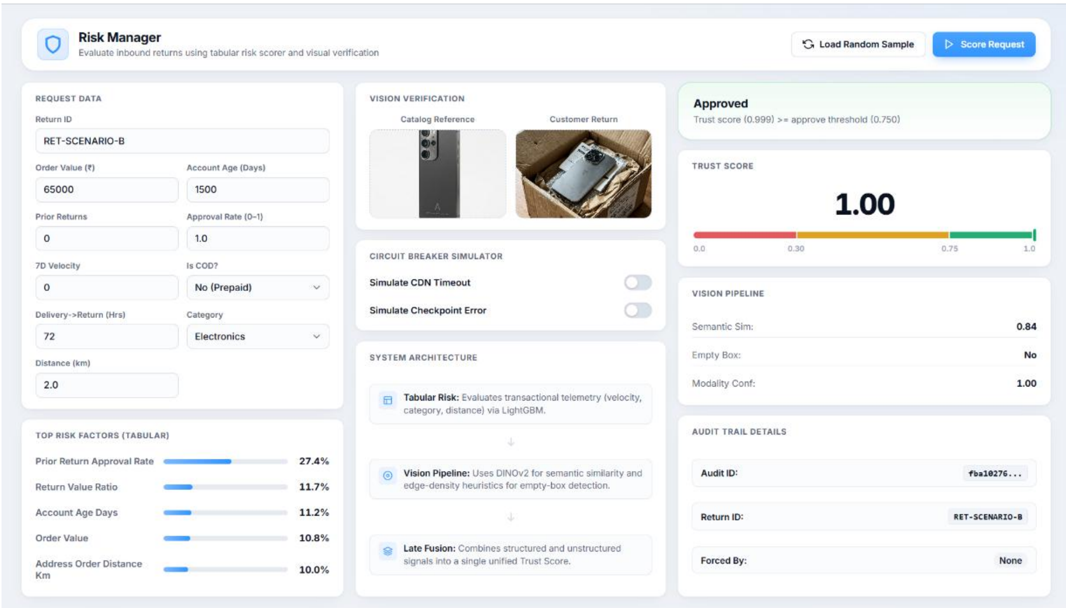
| Component | Implementation detail |
|---|---|
| Tabular Risk Scorer | LightGBM trained on synthetic order/return metadata (return velocity, COD ratio, account age, prior approval rate, address-order distance, delivery-to-return time). |
| Vision - Semantic Similarity | DINOv2 (ViT-S/14) CLS-token embeddings; cosine similarity between the catalog reference photo and the customer's return photo. |
| Vision - Empty-Box Detection | An edge-density / mask-area heuristic (documented as a CPU-friendly fallback for a full SAM2 segmentation pipeline, which was scoped out due to CPU inference-time constraints for a live demo). |
| Late-Fusion Meta-Learner | Logistic Regression combining the tabular score, semantic similarity, empty-box flag, and a modality-confidence term into a single 0-1 Trust Score. |
| Three-Way Decision Router | Auto-Approve (trust $\geq 0.75$) / Nudge - store credit or request another photo (capped at 2 requests) / Manual Review - triggered when trust is low, when a photo is missing, or when the vision pipeline fails (fails safe, never auto-approves on failure). |
| Circuit Breaker | Two independently tested failure modes - image-upload timeout, and a simulated model-checkpoint / embedding-dimension mismatch - both fall back to tabular-only scoring and force a conservative decision rather than crashing or silently degrading. |
| Audit Trail | Every decision, score, threshold version, and failure event is written to an append-only SQLite log, so any decision can be explained after the fact. |
| API & Demo UI | A FastAPI backend exposing a scoring endpoint and failure-injection endpoints, with a single-page dashboard for live demonstration. |

# 7. Results & Live Demonstration:

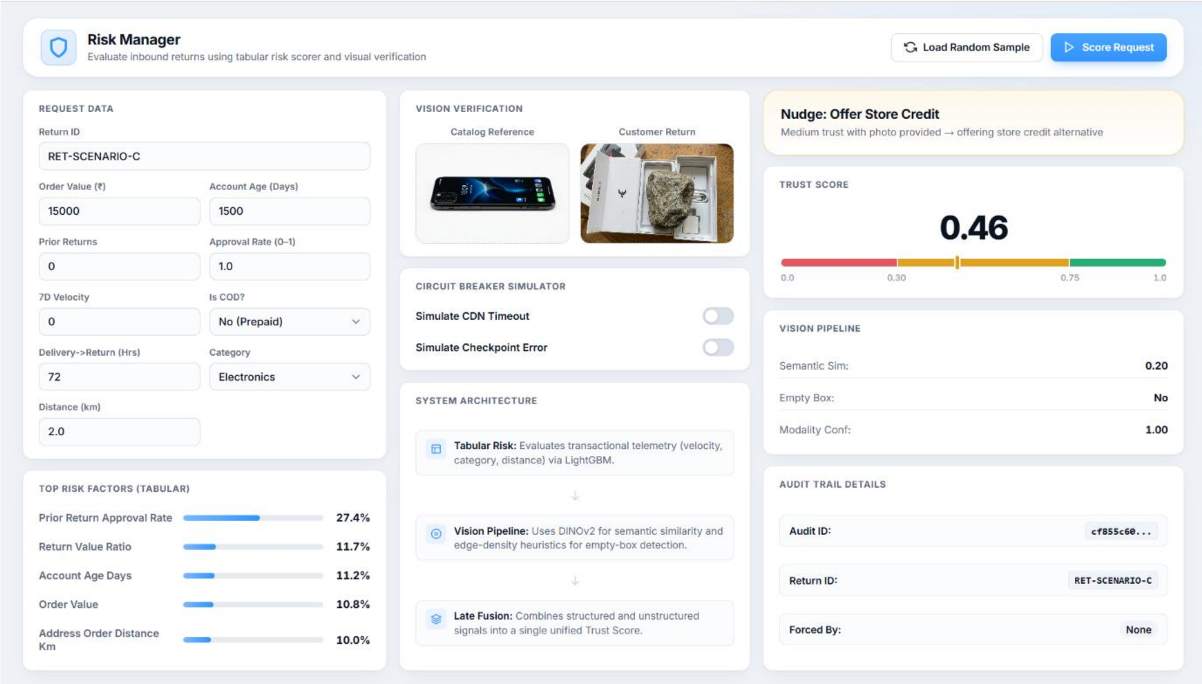
The system was tested live against four representative scenarios, shown below exactly as they rendered in the working demo UI.



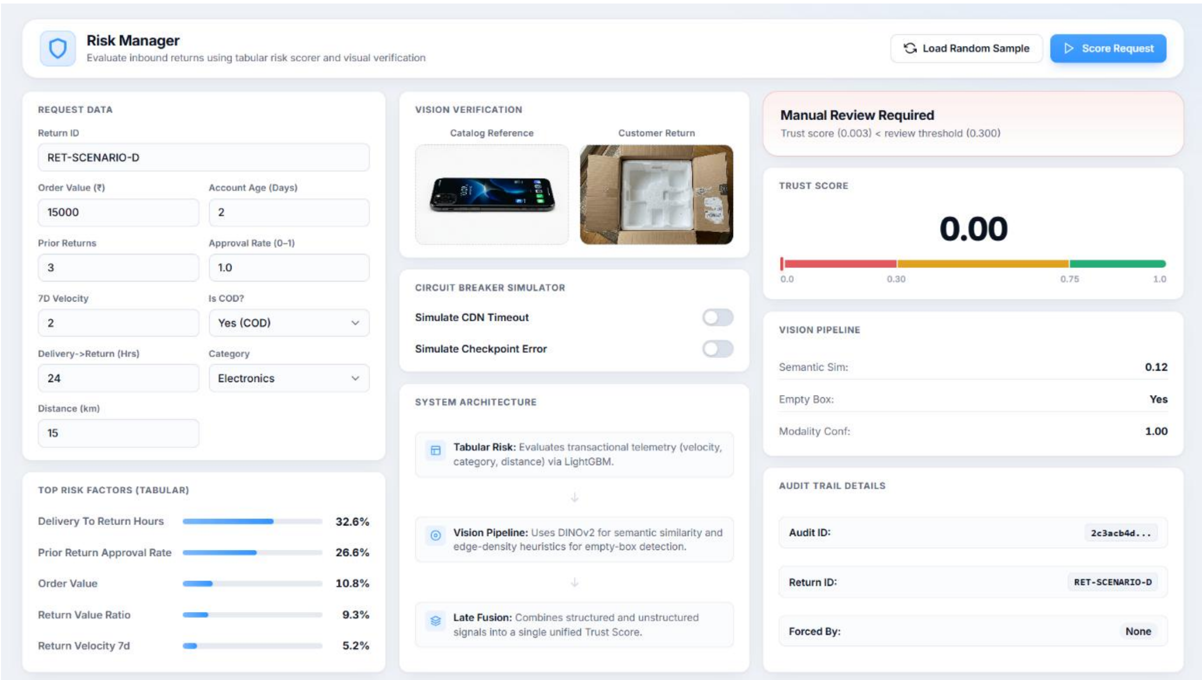

***Scenario A - Auto-Approve (legitimate low-value return).*** *A highly trusted customer returns a pair of headphones. The vision pipeline confirms a strong semantic match (0.81), and the trust score reaches a perfect 1.00, correctly routing to instant approval.*




***Scenario B - Auto-Approve (legitimate high-value return).*** *A trusted customer returns an expensive smartphone. A flawless semantic match (0.84) safely overrides the high-value risk, keeping the trust score at a perfect 1.00 for instant approval.*




***Scenario C - Nudge (substitution fraud).*** *A trusted customer attempts to return a rock in place of a smartphone. While their account history is excellent, semantic similarity collapses to 0.20 and trust falls to 0.46, safely downgrading the request to a store-credit nudge.*




***Scenario D - Manual Review (empty box fraud).*** *A suspicious, high-risk account claims they received an empty box. The vision pipeline edge-density heuristic confirms the box is empty, plummeting the trust score to an absolute 0.00 and correctly forcing a manual review.*

## 8. Honest Metrics & Validity Boundaries:

| Metric | Value |
|---|---|
| Tabular-only PR-AUC | 0.4466 |
| Fusion PR-AUC | 0.8761 |
| Cost-optimal threshold (tabular) | 0.783 (Precision 0.624, Recall 0.331) |
| Cost-optimal threshold (fusion) | 0.778 (Precision: 0.879, Recall: 0.769) |
| Decision distribution (test set) | Auto-approve: 71.8% / Nudge: 19.9% / Manual review: 8.4% |

**Note on Thresholds:** While the mathematical cost-optimal threshold for this specific dataset is 0.778, the live system defaults to a slightly more permissive threshold of 0.75 to prioritize customer experience. Merchants can adjust this via the configuration file.

**Disclosure on the Fusion PR-AUC.** Empirical testing reveals that raw DINOv2 similarity is highly sensitive to the lighting and background differences between clean catalog images and messy real-world return photos. Because of this, the raw cosine similarity occupies a very narrow band. Genuine matches typically score between 0.59 and 0.79, while mismatched items cluster tightly between 0.47 and 0.53. To resolve this compression, we implemented a dynamic scaling mechanism using high and low similarity bounds. This algorithm actively stretches the narrow measurement band across the full zero-to-one range. Consequently, a mismatched item now scores significantly below the neutral baseline, effectively transforming a bad visual match into strong negative evidence. Following this recalibration, our staged dataset yields highly distinct scores: genuine matches span comfortably from 0.394 to 0.840, while fraud mismatches are aggressively isolated between 0.044 and 0.222.

**What these metrics do prove:** The fusion architecture can combine two independently scored signals into a single trust score that responds correctly to real changes in evidence (as shown in Section 7); the cost-weighted threshold methodology produces a defensible, non-arbitrary operating point; and the system degrades to a documented, safe floor (tabular-only performance) when vision is unavailable.

**What they do not prove:** That these exact precision/recall numbers would hold on real merchant data at scale. The tabular feature distributions are grounded in published industry statistics, not any specific merchant's real data, and the vision pipeline has been validated on a small, controlled, staged image set using simulated end-to-end vision scores rather than noisy real-world customer uploads.

## 9. Uniqueness & Differentiation:

- Two independently verifiable signals, not one. Most single-week hackathon entries for this brief will submit a tabular classifier alone; this system fuses behavioural and visual evidence, and the fusion is a real architectural choice (late fusion) rather than a concatenation of features.
- A cost function, not a leaderboard metric. The operating threshold is chosen to minimise estimated business cost, with the cost assumptions stated explicitly rather than hidden inside an F1-optimal default.
- A demonstrated failure, not a narrated one. Both circuit-breaker failure modes were actually triggered and captured on video / in logs, including the real database rows written when each failure occurred - rather than a README claiming failure-handling exists.
- A nudge, not a block. Low-trust cases are not auto-rejected; they are offered store credit or a second photo request (capped to prevent an information-leak loop), which limits merchant payout risk while remaining strictly defense-only.
- Caught and disclosed its own weakest claim. During development, the fusion PR-AUC was found to rely on simulated rather than real end-to-end vision scores. Rather than quietly keeping the headline number, this was investigated, confirmed, and disclosed prominently in both the README and this report (Section 8) - a deliberate choice to be defensible under technical scrutiny rather than to look better on paper.

## 10. Limitations & Future Work:

- AI-generated fake damage photos are the fastest-growing return-fraud vector reported in 2026, and this system does not address it. Building a detector for AI-generated forgeries would require creating synthetic forged images as training data - this constitutes offense-capable tooling and is explicitly disqualified by the competition brief. This is scoped as future work contingent on access to a pre-existing, ethically sourced forgery dataset.

- Scale of real vision data is the primary gap versus production systems: recalibrating the vision pipeline and computing a genuine end-to-end fusion PR-AUC would require real customer return photos at a scale this solo, zero-budget build could not generate.

- Device/identity intelligence (used by platforms such as Sift, Signifyd, and Riskified) was intentionally out of scope, as it requires infrastructure beyond a single buildathon submission.
- SAM2 segmentation was documented as the production-grade approach for empty-box detection, with a CPU-friendly edge-density heuristic used for the live demo after profiling showed SAM2 inference time was impractical for a real-time demo on CPU-only hardware.

## 11. Conclusion:

This submission implements a production-grade architecture validated at a demo-appropriate scale: a late-fusion system combining a tabular risk scorer and a real, independently-verified computer-vision pipeline, routed through a cost-aware three-way decision layer, with two genuinely demonstrated failure-recovery paths and a complete audit trail. It satisfies every stated requirement of Track 02 - one bounded loss class, a working detector/verifier/auto-responder, measured precision and recall on a held-out test set, honest metrics including false-positive cost, and a strictly defense-only design - and it discloses its own evaluation limitations explicitly rather than overstating them.

The intent behind this project was not to claim production readiness, but to demonstrate the engineering judgment that production readiness requires knowing what a system can prove, what it cannot yet prove, and building the parts - graceful degradation, auditability, and honest cost accounting - that most hackathon-scale submissions skip.